Unit V: Plant Physiology (Functional Organisation)

Chapter 15

Plant Growth and Development

Learning Objectives

The learner will be able to,

- *Define growth.*
- *List out and differentiate the phases of growth.*
- *Explain the structure, precursor, bioassay and physiological effects of plant growth regulators.*

Chapter Outline

G2M1R

15.1 Characteristics of growth
15.2 Plant growth regulators
15.3 Plant movements
15.4 Photoperiodism
15.5 Vernalization
15.6 Seed germination and dormancy
15.7 Senescence
15.8 Stress physiology

The Banyan tree continues to grow for thousands of years and some others particularly annual plants cease growth within a season or within a year. Can you understand the reasons? How does a zygote give rise to an embryo and an embryo to a seedling? How does a new plant structure arise from the pre-existing structure? Growth is defined as an irreversible permanent increase in size, shape, number,volume and dry weight. Plant growth occurs by cell division, cell enlargement, differentiation and maturation.

DO YOU KNOW?

Growth is measurable, it is amazing to know that one single maize root apical meristem can give rise to more than 17,500 new cells per hour and cells in a watermelon may increase in size upto 3,50,000 times.

DO YOU KNOW?

Bamboos are evergreen grasses and certain species of it can grow at the rate of growth 91 cm per day. The Saguaro Cactus is a tree like cactus and is a slow growing plant. The rate of growth is one inch in the first ten years and it does not begin to flower until it is about 60 years old. It's lifespan exceeds 150 years and takes 75–100 years to grow a side arm.





15.1 Characteristics of Growth

- Growth increases in protoplasm at cellular level.
- Stem and roots are indeterminate in growth due to continuous cell division and is called **open form of growth**.
- The primary growth of the plant is due to the activity of apical meristem where, new cells are added to root and shoot apex causing linear growth of plant body.
- The secondary vascular cambium and cork cambium add new cells to cause increase in girth.
- Leaves, flowers and fruits are limited in growth or determinate or **closed form growth.**
- Monocarpic annual plants produce flowers only once during lifetime and dies. Example: Paddy and Bean

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